



CLOUD COMPUTING

Introduction to Cloud Computing Terminologies Parallel, Distributed

Dr. Prafullata Kiran Auradkar

Department of Computer Science and Engineering

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Cloud Computing

- Cloud computing is basically delivering computing at the Internet scale.(T2)
- Compute, storage, networking infrastructure as well as development and deployment platforms are made available on-demand within minutes(T2)
- *Cloud computing is a model for enabling **ubiquitous (where-ever)**, convenient, **on-demand (when-ever)** network access to a **shared pool** of configurable computing resources (e.g., networks, servers, storage, applications, and services) that may be **shared** (across applications) and can be **rapidly provisioned and released (quickly)** with **minimal management effort** or service provider interaction*
- **Data centers** house the physical compute, storage and networking infrastructure (along with components which enable functioning or running of these active IT components). These Data center's are also called *Cloud Data center's* when they host these physical resources and enable Cloud computing over the Internet.
- Large clouds, predominant today, often have functions distributed over multiple locations from central servers
- If the connection to the user is relatively close, it may be designated an edge server.
(https://en.wikipedia.org/wiki/Cloud_computing)

Features/Characteristics of Cloud Computing(T2)

On demand self-service: The compute, storage or platform resources needed by the user of a cloud platform are self-provisioned or auto provisioned with minimal configuration.

Broad network access: Ubiquitous access to cloud applications from desktops, laptops to mobile devices is critical to the success of a Cloud platform

Resource pooling: Cloud services can support millions of concurrent users; cloud services need to share resources between users and clients in order to reduce costs.

Scalability: Cloud Services can accommodate larger or smaller loads while supporting some of the expectations of QoS like response time

Rapid elasticity: A cloud platform should be able to rapidly increase or decrease computing resources as needed.

Measured service: One of the compelling business use cases for cloud computing is the ability to “**pay as you go,**” where the consumer pays only for the resources that are actually used by their applications

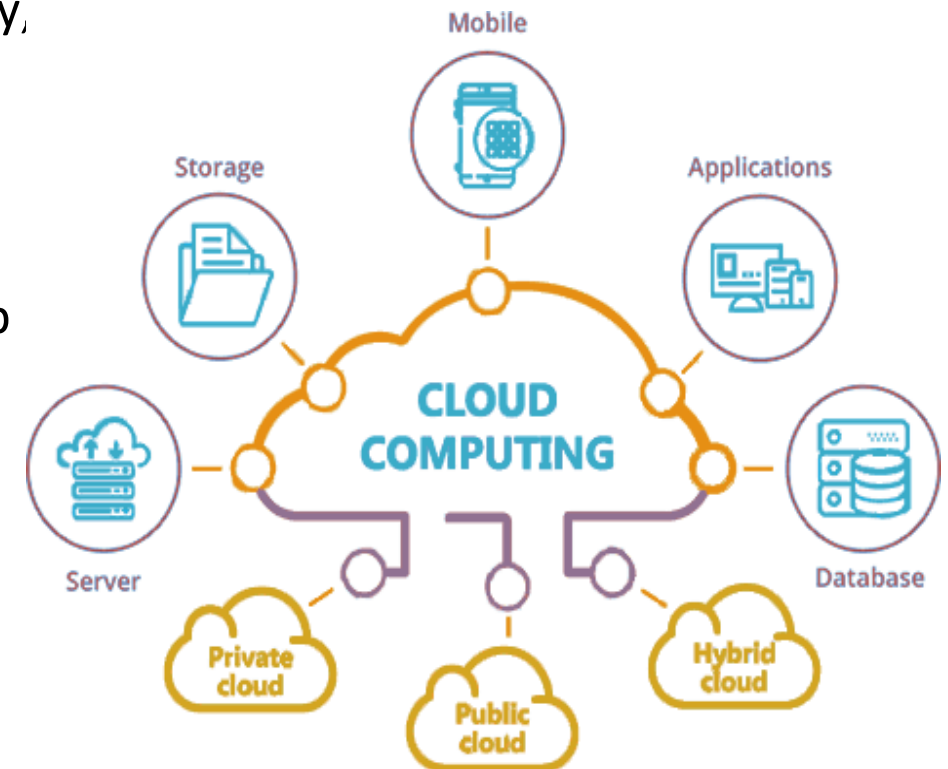
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Introduction to Cloud Computing

Cloud computing usage (<https://aws.amazon.com/what-is-cloud-computing/>)

Organizations of every type, size, and industry are using the cloud for a wide variety of use cases, such as data backup, disaster recovery, email, virtual desktops, software development and testing, big data analytics, and customer-facing web applications.

- For example, healthcare companies are using the cloud to develop more personalized treatments for patients.
- Financial services companies are using the cloud to power real-time fraud detection and prevention.
- And video game makers are using the cloud to deliver online games to millions of players around the world.



Agility

Quickly spin up resources as you need them—from infrastructure services, such as compute, storage, and databases, to Internet of Things, machine learning, data lakes and analytics, and much more.

Elasticity

Scale these resources up or down to instantly grow and shrink capacity as your business needs change.

Cost savings

The cloud allows you to trade capital expenses (such as data centers and physical servers) for variable expenses, and only pay for IT as you consume it.

Deploy globally in minutes

With the cloud, you can expand to new geographic regions and deploy globally in minutes

- Over the past 60 years, computing technology has undergone a series of platform and environment changes
- There have been number of changes in machine architecture, operating system platform, network connectivity, and application workload.
- We have evolved from the use of a centralized computer to solve computational problems, to the use of parallel and distributed computing with multiple computers to solve large-scale problems over the Internet.
- The general computing trend is to leverage shared web resources and massive amounts of data over the Internet.

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Evolution of Cloud Computing (T1 1.1.1.1)

- Billions of users use the internet everyday and need to be provisioned concurrently with High performance computing capability to a large number of users
- High performance computers (HPC) providing this capability for short amounts of time is no longer optimal.
- High throughput computing (HTC) is what is needed using distributed parallel computing.
- Since 1990, we see the proliferation and the use of both HPC and HTC systems hidden in clusters, grids, or Internet clouds.
- These systems are employed by both consumers and high-end workstations for computing and information services
- The general computing trend is to leverage shared web resources and large amounts of data over the Internet

High performance computing (HPC) is the ability to process data and perform complex calculations at high speeds (mostly for short amount of time .. Hours/days)

High Performance Compute (HPC) could also be viewed as a system which has components which can provide high performance to applications using it.

High Throughput Compute (HTC) could be viewed as ability to handle large amounts of computing, for much longer times (months and years, rather than hours and days)

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Evolution of Cloud Computing (T1 1.1.1.1)

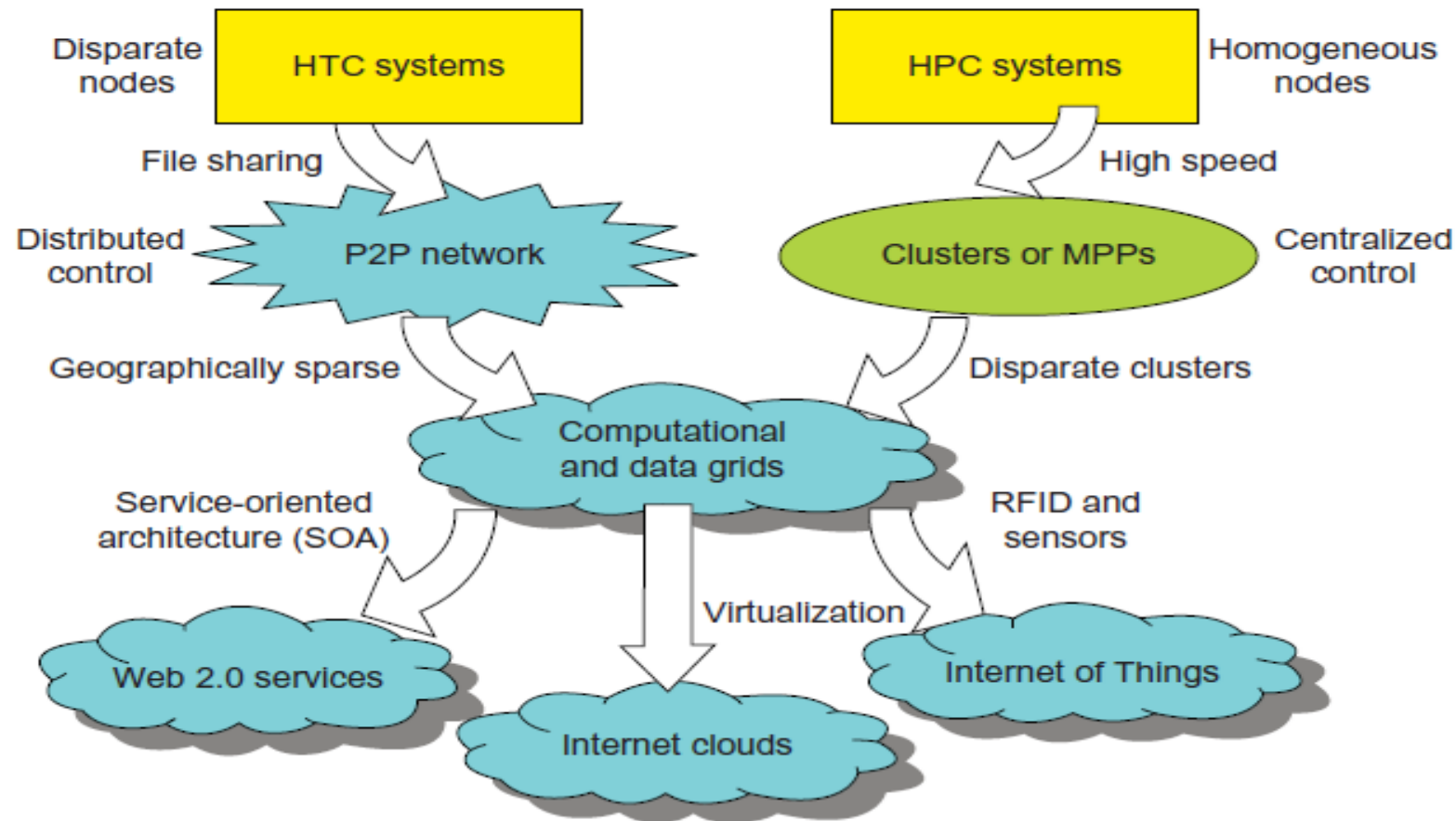


FIGURE 1.1

Evolutionary trend toward parallel, distributed, and cloud computing with clusters, MPPs, P2P networks, grids, clouds, web services, and the Internet of Things.

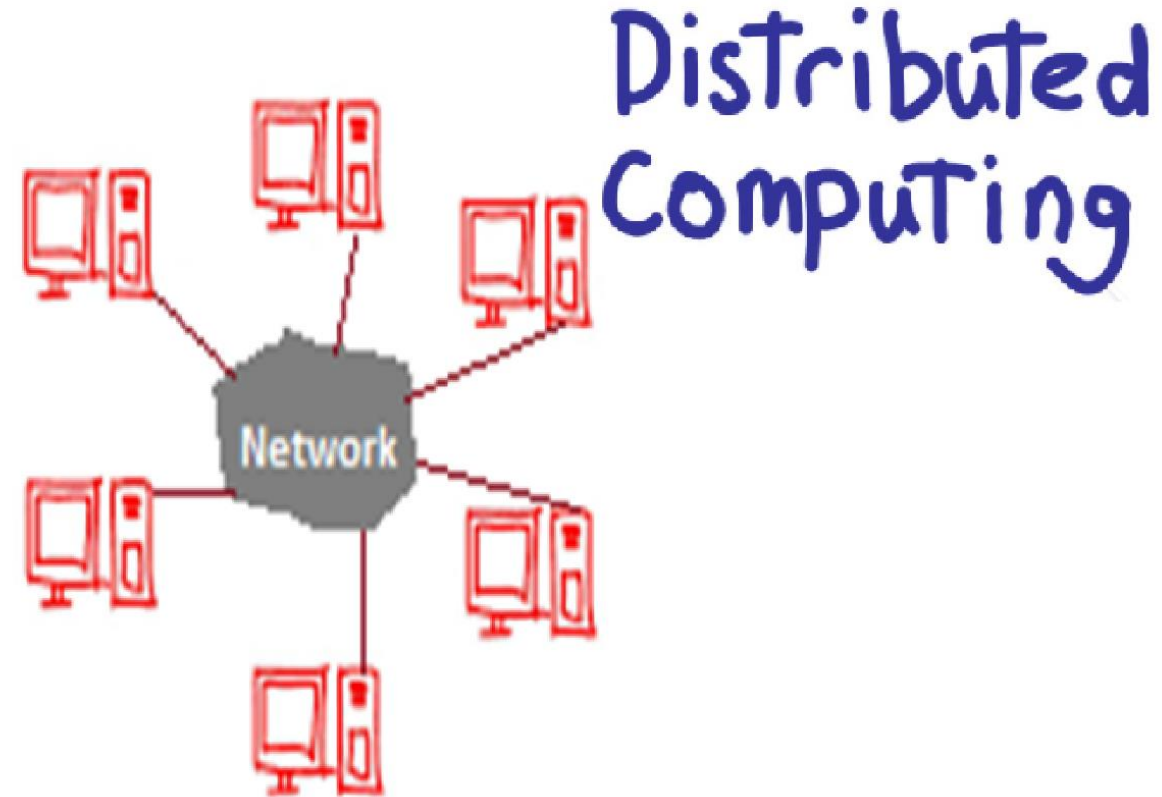
- HPC generally refers to the practice of aggregating computing power in a way that delivers much higher performance than one could get out of a typical desktop computer or workstation in order to solve large problems in science, engineering, or business.
- A helpful way to understand what high performance computers are is to think about the what's in them. You have all of the elements you'd find on your desktop — processors, memory, disk, operating system — just more of them
- High performance computers, which are of interest to small and medium-sized businesses today are really *clusters* of computers.
- People often refer to the individual *computers* in a cluster as *nodes*
- HPC systems are mostly expected to support for short amount of time .. In Hours/days

Distributed computing (multiple complete systems)

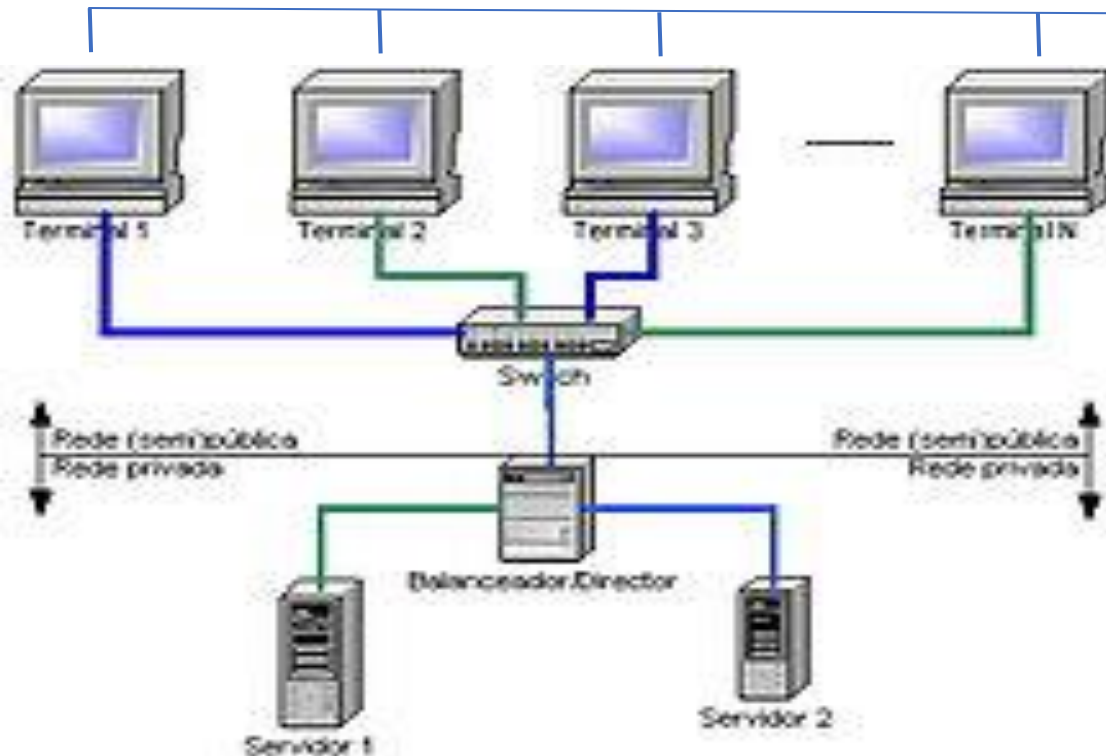
A distributed system consists of multiple autonomous computers, each having its own private memory, communicating through a computer network. Information exchange in a distributed system is accomplished through message passing

Distributed system components can be located on different networked computers that coordinate their actions by communicating via pure HTTP, RPC-like connectors or message queues.

Significant characteristics of distributed systems include independent failure of components and concurrency of components



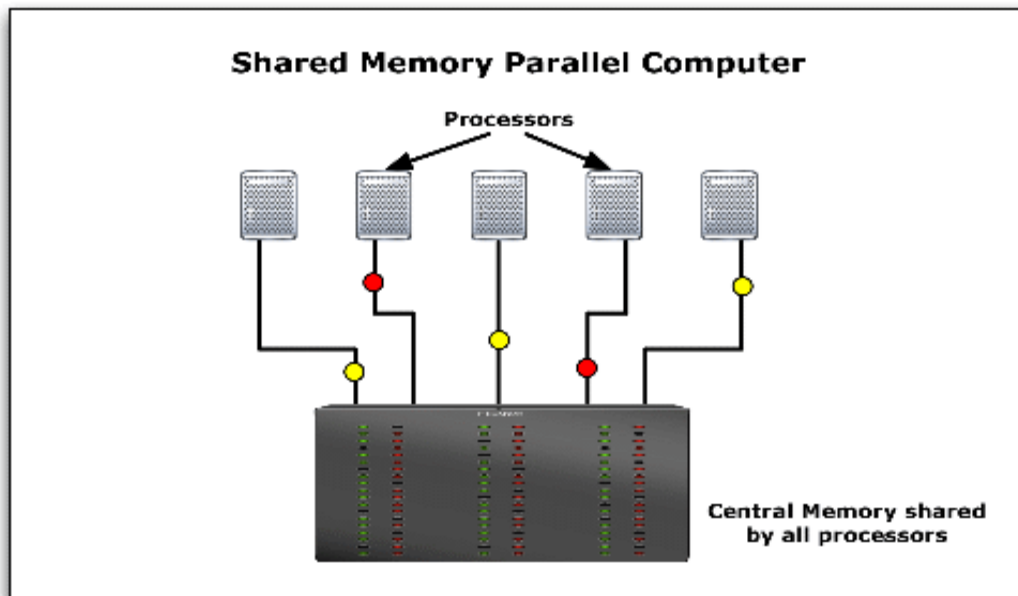
A computer cluster is a set of loosely or tightly connected computers that work together so that, in many aspects, they can be viewed as a single system. Clusters have each node set to perform the same task, controlled and scheduled by software.



Mostly have similar hardware configuration and run the same OS (different instances though)

Parallel Computing (Single system with multiple processors)

In parallel computing, all processors are either tightly coupled with centralized shared memory or loosely coupled with distributed memory. Inter-processor communication is accomplished through shared memory or via message passing.



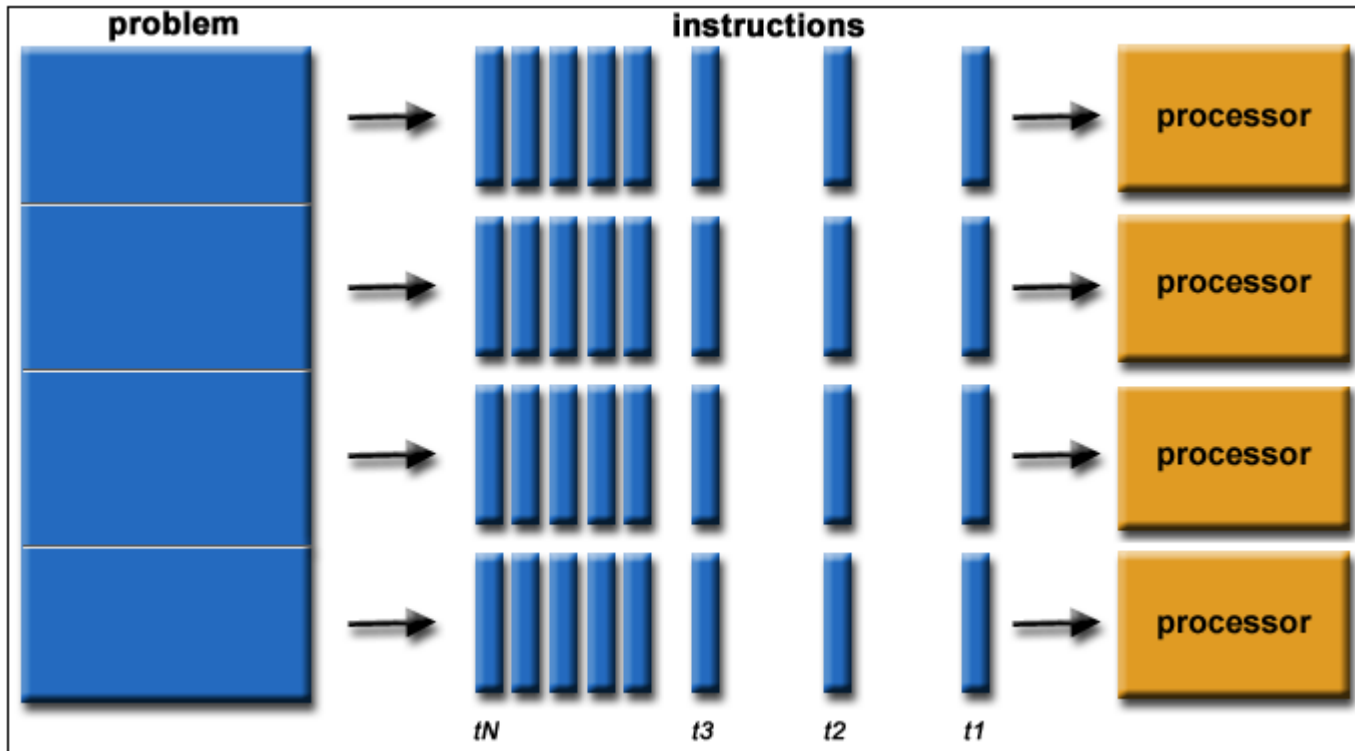
The primary goal of parallel computing is **to increase available computation power** for faster application processing and problem solving

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Evolution of Computing : Parallel Computing

```
Last login: Tue Jan 10 11:46:58 2023 from 10.10.3.60
student1@v2g2:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Byte Order:             Little Endian
CPU(s):                 32
On-line CPU(s) list:   0-31
Thread(s) per core:    2
Core(s) per socket:    16
Socket(s):              1
NUMA node(s):          4
Vendor ID:              AuthenticAMD
CPU family:             23
Model:                  1
Model name:             AMD EPYC 7301 16-Core Processor
Stepping:               2
CPU MHz:                2694.455
CPU max MHz:            2200.0000
CPU min MHz:            1200.0000
BogoMIPS:               4399.32
Virtualization:         AMD-V
L1d cache:              32K
L1i cache:              64K
L2 cache:               512K
L3 cache:               8192K
NUMA node0 CPU(s):     0-3,16-19
NUMA node1 CPU(s):     4-7,20-23
NUMA node2 CPU(s):     8-11,24-27
NUMA node3 CPU(s):     12-15,28-31
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cm
ov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdt
scp lm constant_tsc rep_good nopl nonstop_tsc cpuid extd_apicid amd_dcm aperfmpe
rf pni pclmulqdq monitor ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx
f16c rdrand lahf_lm cmp_legacy svm extapic cr8_legacy abm sse4a misalignsse 3dn
owprefetch osvw skinit wdt tce topoext perfctr_core perfctr_nb bpext perfctr_llc
mwaitx cpb hw_pstate ssbd ibpb vmcall fsgsbase bmi1 avx2 smep bmi2 rdseed adx
smap clflushopt sha_ni xsaveopt xsavec xgetbv1 xsaves clzero irperf xsaveerptr a
rat npt lbrv svm_lock nrip_save tsc_scale vmcb_clean flushbyasid decodeassists p
ausefilter pfthreshold avic v_vmsave_vmload vgif overflow_recov succor smca
```

In Parallel Computing several processors simultaneously execute multiple, smaller calculations broken down from an overall larger, complex problem. These smaller calculations communicate through shared memory and the results of these can be combined as part of an overall program



There are various degrees of parallelism

- Bit Level Parallelism
- Instruction Level Parallelism
- Task Level Parallelism
- Data Parallelism

Bit Level Parallelism (<https://www.tutorialspoint.com/types-of-parallelism-in-processing-execution>)

Bit-level parallelism is a form of parallel computing which is based on **increasing processor word size**. In this type of parallelism, with increasing the word size **reduces the number of instructions** the processor must execute in order to perform an operation on variables whose sizes are greater than the length of the word.

E.g., consider a case where an 8-bit processor must add two 16-bit integers. First the 8 lower-order bits from each integer were added by processor, then add the 8 higher-order bits, and then two instructions to complete a single operation. A processor with 16-bit would be able to complete the operation with single instruction.

Instruction Level Parallelism

A processor can only address less than one instruction for each clock cycle phase. These instructions can be re-ordered and grouped which are later on executed concurrently without affecting the result of the program. So we achieve concurrent execution of multiple instructions from a program. While pipelining is a form of ILP, we must exploit it to achieve parallel execution of the instructions in the instruction stream.

Example

```
for (i=1; i<=100; i= i+1)
    y[i] = y[i] + x[i];
```

This is a parallel loop. Every iteration of the loop can overlap with any other iteration, although within each loop iteration there is little opportunity for overlap.

Data Parallelism

Data Parallelism means concurrent execution of the same task on each multiple computing core.

Let's take an example, summing the contents of an array of size N . For a single-core system, one thread would simply sum the elements $[0] \dots [N - 1]$. For a dual-core system, however, thread A, running on core 0, could sum the elements $[0] \dots [N/2 - 1]$ and while thread B, running on core 1, could sum the elements $[N/2] \dots [N - 1]$. So the Two threads would be running in parallel on separate computing cores.

Task Parallelism

Task Parallelism means concurrent execution of the different task on multiple computing cores.

Consider again our example above, an example of task parallelism might involve two threads, each performing a unique statistical operation on the array of elements. Again The threads are operating in parallel on separate computing cores, but each is performing a unique operation.

Parallel Computing Architecture

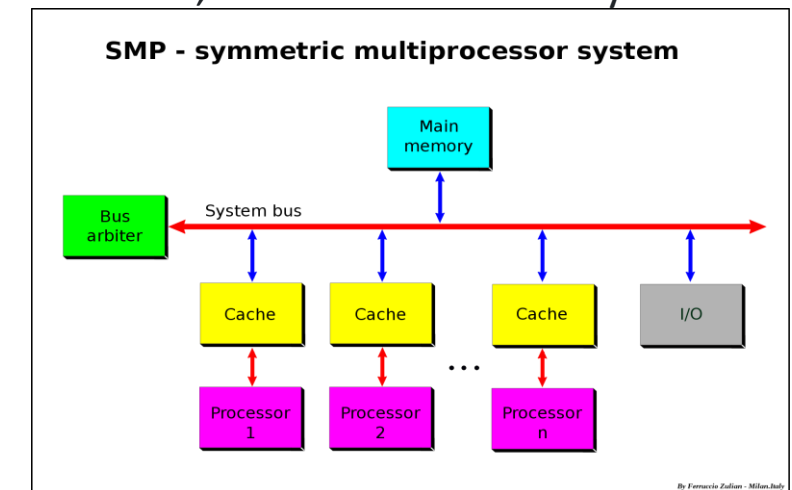
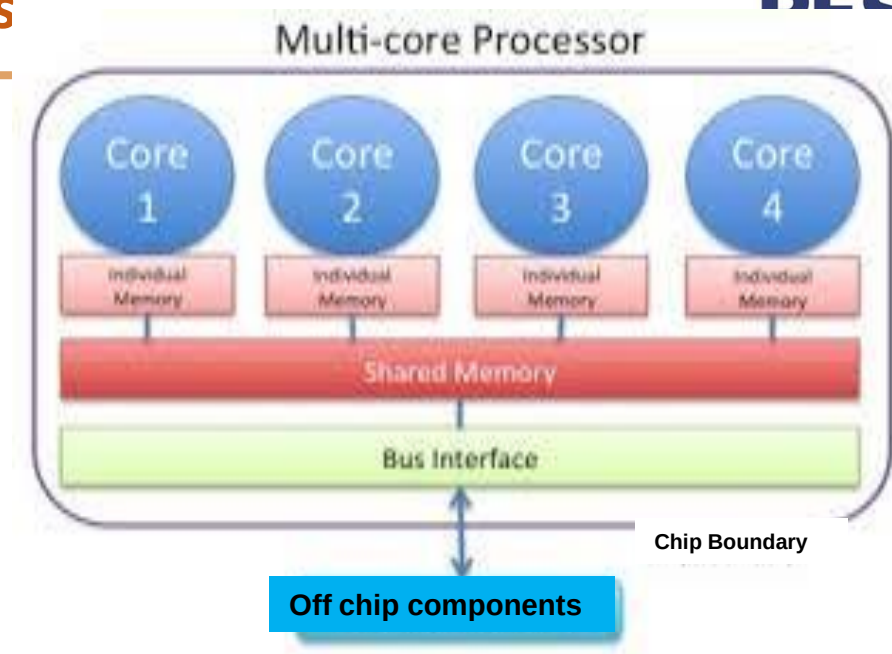
Multi-core computing: A multi-core processor is a computer processor integrated circuit with two or more separate processing cores, each of which executes program instructions in parallel.

Symmetric multiprocessing: multiprocessor computer hardware and software architecture in which two or more independent, homogeneous processors are controlled by a **single operating system** instance that treats all processors equally, and is connected to a **single, shared main memory** with full access to all common resources and devices

Each processor has a private cache memory, may be connected using on-chip mesh networks, and can work on any task no matter where the data for that task is located in memory.

Massively parallel computing: refers to the use of numerous computers or computer processors to simultaneously execute a set of computations in parallel.

One approach involves the grouping of several processors in a tightly structured, centralized computer cluster. Another approach is grid computing, in which many widely distributed computers work together and communicate via the Internet to solve a particular problem.



Parallel Computing Software Solutions and Techniques

- **Application checkpointing:** a technique that provides fault tolerance for computing systems by recording all of the application's current variable states, enabling the application to restore and restart from that point in the instance of failure.
- **Automatic parallelization:** refers to the conversion of sequential code into multi-threaded code in order to use multiple processors simultaneously in a shared-memory multiprocessor (SMP) machine.
- **Parallel programming languages:** Parallel programming languages are typically classified as either distributed memory or shared memory. While distributed memory programming languages use message passing to communicate, shared memory programming languages communicate by manipulating shared memory variables.



THANK YOU

Prafullata Kiran Auradkar

Department of Computer Science and Engineering

prafullatak@pes.edu